



Autonomous Agents on the Web (WebAgents) Community Group

Co-chairs: Andrei Ciortea, Rem Collier, Ege Korkan, and Antoine Zimmermann

JULY 17, 2026



Basic Info

Group homepage: <https://www.w3.org/community/webagents/>

Charter: <https://www.w3.org/community/webagents/charter/>

GitHub: <https://github.com/w3c-cg/webagents/>

Calendar: <https://www.w3.org/groups/cg/webagents/calendar/>

Goals

This CG aims to investigate the design of a new class of Web-based Multi-Agent Systems (MAS) that:

- inherit the beneficial architectural properties of the Web (Internet-scalability, evolvability, simplicity, etc.),
- preserve the beneficial properties of MAS (adaptability, openness, robustness, etc.), and
- are human-centric (support transparency, usability, accountability, etc.).

Agents on the Web Community



May 2019 • HyperAgents 2019 @ TheWebConf 2019



Dagstuhl Seminar 21072:
“Autonomous Agents on the Web”

Feb. 2021



<https://www.dagstuhl.de/21072>

Feb. 2023

Dagstuhl Seminar 23081:
“Agents on the Web”



W3C WebAgents
Community Group

March 2023



<https://www.dagstuhl.de/23081>

<https://www.w3.org/community/webagents/>



Activity Overview

Three Task Forces:

- Use Cases
- Manageable Affordances
- Interoperability
- TBA: Interaction

Hybrid Meetings: TPAC 2023, EUMAS 2024, WoT Week 2024, ECAI 2025, TPAC 2025

- TPAC 2026: October 26, 09:00 – 12:30 GMT

Interoperability Task Force





W3C Community Group
Draft Report

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- 4. Identification**
 - 4.1 Relevant Standards and Initiatives
 - 4.1.1 Agent Identification

Latest published version:
none

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<https://w3c-cg.github.io/webagents/TaskForces/Interoperability/Reports/report-interoperability.html>

Editors:
[Andrei Ciortea](#)  (Inria and University of St.Gallen)
[Rem Collier](#)  (University College Dublin)

Authors:
[Jérémy Lemée](#)  (University of St.Gallen)
[Danai Vachtsevanou](#)  (University of St.Gallen)
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Feedback:
[GitHub w3c-cg/webagents](#) ([pull requests](#), [new issue](#), [open issues](#))

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Abstract

Advances in large language models (LLMs) that can follow instructions and interact with the web have enabled the emergence of autonomous agents and multi-agent systems. Like previous generations of AI, these systems are designed for specific tasks, highlighting the need for open networks of agents and protocols that enable their abilities to tackle more complex problems. New protocols are rapidly emerging...



<https://bit.ly/webagents-interop>

What should be the **role of the Web** in the emerging landscape of Agentic AI?

What are the **relevant standards** within and maybe also outside the W3C?

What are the **standardization gaps** (if any)?

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Is It an Agent, or Just a Program?: A Taxonomy for Autonomous Agents

Stan Franklin and Art Graesser

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Abstract

The advent of software agents gave rise to much discussion of just what such an agent is, and of how they differ from programs in general. Here we propose a formal definition of an autonomous agent which clearly distinguishes a software agent from just any program. We also offer the beginnings of a natural kinds taxonomy of autonomous agents, and discuss possibilities for further classification. Finally, we discuss subagents and multiagent systems.

1 Introduction

On meeting a friend or colleague that we haven't seen for a while, or a new acquaintance, some version of the following conversation often ensues:



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§ 2. Terminology

Agent or Autonomous Agent

An entity [situated](#) in an environment that perceives its environment and acts on it, over time, in pursuit of its goals. For a detailed discussion of agent definitions, see [\[FRANKLIN96\]](#).

Agent Interaction Protocol

A specification of communication among two or more [agents](#) that states who can say what to whom and when — for example, as message sequence diagrams [\[AUML\]](#) or information flows [\[BSPL\]](#).

Augmented Language Model

A language model augmented with abilities such as reasoning, tool use, information retrieval, or storing context across interactions. Unlike an [agent](#), an augmented language model does not actively pursue goals and is not [situated](#) in an environment. See also [\[TMLR23\]](#) and [\[ANTHROPIC24\]](#).

LLM Agent or Language Agent

An [agent](#) that relies on an LLM to guide their internal processes and interactions with the environment, while maintaining control over how they accomplish tasks [\[ANTHROPIC24\]](#)[\[COALA23\]](#). [This is the sort of agent people think about when they talk about Agentic AI.]

Multi-Agent System (MAS)

A system composed of [agents](#) that are situated in a shared environment and interact with one another to achieve individual or collective goals. Agents can work in collaboration, cooperation, and/or competition. A [MAS](#) can be either an open or a closed system. This report is primarily concerned with open [MAS](#).

Situatedness

The ability of an [agent](#) to interact with its environment directly through perception and action, and to respond in a timely fashion to sensory input.

Tool or Artifact

An instrument that can be shared and used by [agents](#) to support their activities. In some [multi-agent systems](#), agents construct artifacts to instrument their environments [\[JACAMO20\]](#). In the context of agentic

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	Relevant Concepts	Agent Interaction	Tool Use	Identifiers	Descriptions	Discovery Mechanisms	Arch. Style
MCP	Tool , Resource , Prompt	N/A	Function calling	Strings (Tools and Prompts), URIs (Resources)	Tool definition , Resource descriptions , Prompt definitions , (JSON)	Directories (via */list)	Client-Server with streaming RPC connectors (JSON-RPC 2.0, Streamable HTTP)
A2A	Agent Card , Task	Task invocation	N/A	Strings?	Agent Card , Task description , (JSON)	Well-known URIs, Directories	Async. Client-Server with streaming RPC connectors and webhooks (JSON-RPC 2.0, HTTP+SSE)
ANP	Agent , Agent Description , Communication Protocol	Communication protocols with protocol negotiation	N/A	W3C DID with custom Web-based Agent DID Method	Agent Description (RDF/JSON-LD)	Directories	Peer-to-Peer? (WebSocket subprotocol)
LMOS	Agent , Agent Group , Tool , Agent Description , Tool Description	Message passing? (in principle: TD interaction affordances)	Property Affordances, Event Affordances, Action Affordances (W3C WoT TD)	Uniform identifiers (IRIs, W3C DIDs)	Agent Description , Tool Description (W3C WoT TD; JSON, RDF/JSON-LD)	DNS-SD/mDNS, Well-known URIs, Directories (W3C WoT Discovery)	W3C WoT Arch.? with protocol bindings for HTTP and WebSocket subprotocol

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Feedback:
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Abstract

Advances in large language models (LLMs) that can follow instructions have enabled the design of autonomous agents and multi-agent systems. Like previous systems, these agents are designed for specific tasks, highlighting the need for open standards and protocols that enable agents to tackle more complex problems. New protocols are needed to enable agents to interact with the Web and to leverage the Web's capabilities to tackle more complex problems. New protocols are needed to enable agents to interact with the Web and to leverage the Web's capabilities to tackle more complex problems.

What goals should guide the design of Web-based Multi-Agent Systems (MAS)?

How can the Web contribute to those design goals?

What is a minimal set of architectural patterns for Web-based MAS?

What architectural constraints do we need in order to fully leverage the Web?

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Abstract

Advances in large language models (LLMs), autonomous agents and multi-agent systems designed for specific tasks, highlighting the abilities to tackle more complex problems.

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4. Identification

8. Agent-to-Agent Interaction

5. Profiles

9. Agent-Environment Interaction

6. Verifiable Credentials

10. Policies, Norms, and Accountability

7. Discovery

11. Security and Privacy

12. Conclusions: A Roadmap for Agents on the Web



Thank you!

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